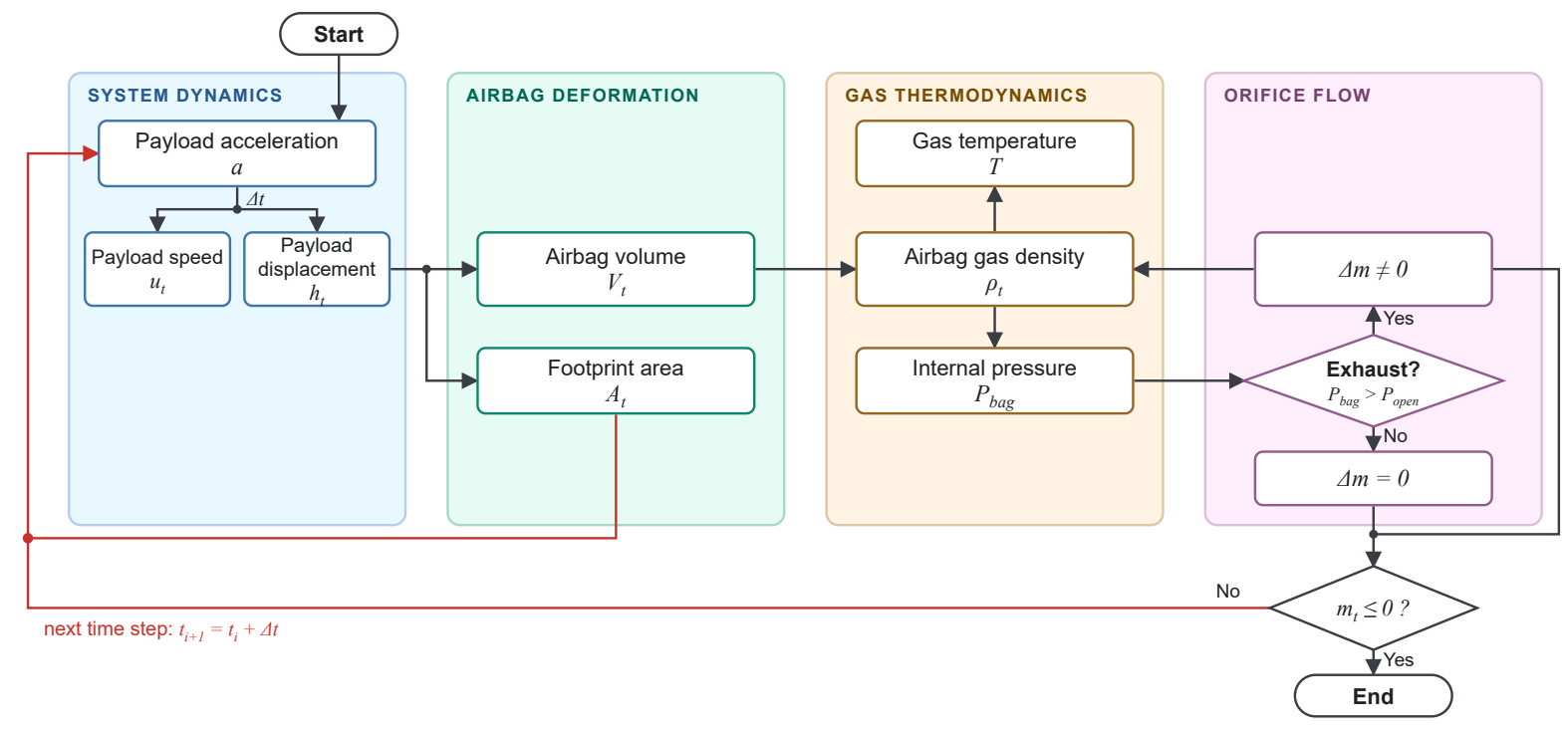

Start
SYSTEM DYNAMICS
Payload acceleration
a
Δt
Payload speed
u_t
Payload displacement
h_t
AIRBAG DEFORMATION
Airbag volume
V_t
Footprint area
A_t
GAS THERMODYNAMICS
Gas temperature
T
Airbag gas density
ρ_t
Internal pressure
P_{bag}
ORIFICE FLOW
$\Delta m \neq 0$
Yes
Exhaust?
$P_{bag} > P_{open}$
No
$\Delta m = 0$
No
$m_t \leq 0$?
Yes
End
next time step: $t_{i+1} = t_i + \Delta t$